Full Short-answer Benchmark

Short-answer questions, reference answers and rubrics

Candidate number: 1102384

This document contains all 55 short-answer benchmark items: 38 supplied by the placement supervisor and 17 developed for this study. It excludes the three extended computational modelling tasks. It is retained as a repository build artefact and is not a submitted supplementary-text file.

1 Basic understanding

SUP-B01 Supervisor-supplied

Question

Why do untreated Plasmodium falciparum parasite counts often fluctuate with a roughly 48-hour periodicity?

Valid reference answer

Because blood-stage parasites progress through a synchronous 48-hour asexual cycle, with maturation and rupture producing waves of circulating parasites.

Rubric (1 criterion)

C1. Explains that the approximately 48-hour pattern arises from the asexual blood-stage developmental cycle and synchronous maturation or rupture.

SUP-B02 Supervisor-supplied

Question

What is sequestration?

Valid reference answer

The movement of mature infected erythrocytes out of peripheral blood into deeper tissues or the microvasculature, reducing what is seen on microscopy.

Rubric (1 criterion)

C1. Defines sequestration as mature infected erythrocytes leaving peripheral circulation for deeper tissue or microvasculature, thereby reducing peripheral microscopy counts.

SUP-B03 Supervisor-supplied

Question

What is an age-structured model?

Valid reference answer

A model that tracks parasites by developmental stage or age rather than as a single undifferentiated total.

Rubric (1 criterion)

C1. Defines an age-structured model as one that tracks parasites by developmental age or stage rather than only as a single total.

SUP-B04 Supervisor-supplied

Question

Why are daily parasite measurements useful?

Valid reference answer

They capture short-term oscillations and changes in growth rate that sparse sampling can miss.

Rubric (1 criterion)

C1. Explains that daily sampling can resolve short-term oscillations or changes in growth that sparser sampling can miss.

SUP-B05 Supervisor-supplied

Question

What is the main difference between microscopy and qPCR parasite measurements?

Valid reference answer

Microscopy counts visible parasites in blood films, whereas qPCR measures parasite nucleic acid and is more sensitive at low densities.

Rubric (2 criteria)

C1. Distinguishes microscopy as visual counting of parasites in blood films from qPCR as measurement of parasite nucleic acid.

C2. Notes that qPCR is generally more sensitive at low parasite densities.

SUP-B06 Supervisor-supplied

Question

Why are observations often censored after antimalarial treatment starts?

Valid reference answer

Treatment changes parasite growth and clearance, so the untreated expansion phase is no longer being observed.

Rubric (1 criterion)

C1. Explains that treatment changes parasite growth or clearance, so post-treatment observations no longer represent untreated expansion.

SUP-B07 Supervisor-supplied

Question

What is the purpose of an observation model?

Valid reference answer

To connect the unobserved latent parasite state to measured data while accounting for measurement error and detection limits.

Rubric (2 criteria)

C1. States that an observation model maps an unobserved biological state to measured data.

C2. Recognises measurement error, detection limits, or related assay properties as part of that mapping.

SUP-B08 Supervisor-supplied

Question

Why can parasite trajectories differ between individuals or strains?

Valid reference answer

Because of differences in synchrony, replication capacity, growth limitation, host environment, and infection history.

Rubric (1 criterion)

C1. Identifies biologically defensible sources of between-individual or between-strain variation, such as synchrony, replication, growth limitation, host environment, or infection history.

SUP-B09 Supervisor-supplied

Question

Why compare models using predictive criteria such as LOO?

Valid reference answer

To assess which model predicts new data better while penalizing unnecessary complexity.

Rubric (2 criteria)

C1. Explains that predictive criteria such as LOO assess expected performance on unseen data.

C2. Recognises that predictive comparison discourages unsupported model complexity.

SUP-B10 Supervisor-supplied

Question

What does it mean to call a model phenomenological?

Valid reference answer

It describes the observed pattern with a few interpretable terms without claiming to represent every underlying biological mechanism explicitly.

Rubric (2 criteria)

C1. Explains that a phenomenological model represents an observed pattern using a compact or interpretable description.

C2. Does not claim that every fitted term necessarily represents a specific underlying biological mechanism.

SUP-B11 Supervisor-supplied

Question

What biological quantity does parasite multiplication factor (PMF) represent in the White et al. model?

Valid reference answer

The average number of merozoites that successfully invade red blood cells per infected erythrocyte completing one cycle.

Rubric (1 criterion)

C1. Defines PMF as the average number of merozoites that successfully establish new infected erythrocytes per infected erythrocyte completing one cycle.

SUP-B12 Supervisor-supplied

Question

Why are only parasites aged 1-26 hours assumed to contribute to observed peripheral blood counts?

Valid reference answer

Parasites older than about 26 hours are assumed to sequester and therefore are not visible in peripheral blood microscopy.

Rubric (1 criterion)

C1. Explains that parasites older than about 26 hours are assumed to sequester and therefore cease contributing to observed peripheral blood counts.

SELF-B01 Self-authored

Question

Which Plasmodium falciparum gametocyte stages are predominantly sequestered, which circulate in peripheral blood, and where does most immature gametocyte development occur?

Valid reference answer

Immature stages I-IV are predominantly sequestered, whereas mature stage-V gametocytes circulate. The extravascular bone marrow is a principal maturation niche.

Rubric (3 criteria)

C1. States that immature gametocyte stages I-IV are predominantly sequestered.

C2. States that mature stage-V gametocytes circulate in peripheral blood.

C3. Identifies extravascular bone marrow as a principal immature-gametocyte maturation site.

SELF-B02 Self-authored

Question

Circulating mature Plasmodium falciparum gametocytes are represented by dG/dt = rG with r > 0. Is this biologically defensible? If not, write a minimal balance equation for G(t).

Valid reference answer

Mature gametocytes do not replicate in human blood. A minimal model is dG/dt = J_V(t) - mu_G G(t), where J_V(t) is the delayed influx of newly matured stage-V gametocytes and mu_G is their clearance or death rate.

Rubric (3 criteria)

C1. Rejects autonomous replication of mature gametocytes in human blood.

C2. Includes an input representing maturation into the circulating stage-V compartment.

C3. Includes a clearance or death loss proportional to the mature-gametocyte compartment.

SELF-B03 Self-authored

Question

A blood-stage infection starts with 10^2 parasites and increases exactly tenfold every 48 hours. With no parasite loss, how many complete cycles and days are required to reach 10^11 parasites?

Valid reference answer

Nine cycles, corresponding to 18 days.

Rubric (2 criteria)

C1. States that nine complete 48-hour cycles are required.

C2. Converts nine 48-hour cycles to 18 days.

SELF-B04 Self-authored

Question

In the cycle model N_(k+1) = M S N_k, M is the parasite multiplication factor and S is the fraction surviving all losses. When does parasite burden decline, and what is the stationary threshold?

Valid reference answer

Burden declines when M S < 1, equivalently S < 1/M. The stationary threshold is M S = 1.

Rubric (2 criteria)

C1. States that parasite burden declines when MS is less than one, equivalently when S is less than 1/M.

C2. States that the stationary threshold is MS equal to one.

2 Applied reasoning

SUP-A01 Supervisor-supplied

Question

What features should a model fitted to microscopy data usually capture?

Valid reference answer

Periodic fluctuations, sequestration-related loss from peripheral blood, timing of growth, and the effect of detection thresholds.

Rubric (2 criteria)

C1. Identifies periodic fluctuations and sequestration-related changes in peripheral visibility.

C2. Includes growth timing or dynamics and the effect of microscopy detection thresholds.

SUP-A02 Supervisor-supplied

Question

What additional features may matter when fitting qPCR data?

Valid reference answer

Lower detection limits, irregular sampling times, and the possibility that the measured signal reflects parasite DNA rather than only visible parasites.

Rubric (2 criteria)

C1. Recognises qPCR's lower detection limit or greater low-density sensitivity.

C2. Recognises irregular observation times or an assay signal based on parasite DNA rather than only microscopically visible parasites.

SUP-A03 Supervisor-supplied

Question

Why might a pure exponential growth model be insufficient for untreated parasite trajectories?

Valid reference answer

Many infections slow down, plateau, or decline rather than growing indefinitely at a constant rate.

Rubric (1 criterion)

C1. Explains that constant exponential growth cannot represent the observed slowing, plateau, or decline of many untreated trajectories.

SUP-A04 Supervisor-supplied

Question

When would you add a density-dependent growth effect to a model?

Valid reference answer

When parasite expansion weakens as density increases or when trajectories show plateauing or decline after an initial rise.

Rubric (1 criterion)

C1. Adds density dependence when expansion weakens as density rises or after an initial rise trajectories plateau or decline.

SUP-A05 Supervisor-supplied

Question

What would explicit sexual commitment add to a within-host model?

Valid reference answer

It would move some parasites out of the asexual replication cycle and allow the model to describe gametocyte production and transmission potential.

Rubric (2 criteria)

C1. Explains that sexual commitment diverts parasites out of the asexual replication cycle.

C2. Connects this diversion to gametocyte production or transmission potential.

SUP-A06 Supervisor-supplied

Question

Why can hierarchical models be helpful when fitting many individual profiles?

Valid reference answer

They share information across individuals, improve estimation of common patterns, and reduce noise in poorly informed profiles.

Rubric (2 criteria)

C1. Explains that hierarchical models share information across individuals.

C2. Connects partial pooling to improved common-pattern estimation or less noisy estimates for poorly informed individuals.

SUP-A07 Supervisor-supplied

Question

Why can individual-level parameter estimates have wide credible intervals?

Valid reference answer

Because each trajectory may contain only a few time points and several parameters can explain the same pattern.

Rubric (1 criterion)

C1. Attributes wide individual credible intervals to sparse observations and/or confounding in which multiple parameter combinations explain the same trajectory.

SUP-A08 Supervisor-supplied

Question

What does overlapping credible intervals imply when comparing strains?

Valid reference answer

The data do not provide strong evidence that the strain-specific differences are clearly separated.

Rubric (1 criterion)

C1. Interprets overlapping credible intervals cautiously as a lack of strong evidence that strain effects are clearly separated.

SUP-A09 Supervisor-supplied

Question

How should you interpret similar LOO performance for two models?

Valid reference answer

The data do not strongly favor one model; both explain the observations about equally well.

Rubric (1 criterion)

C1. States that similar LOO performance provides little evidence that either model predicts better than the other.

SUP-A10 Supervisor-supplied

Question

What would suggest that a more complex model is unnecessary?

Valid reference answer

If it does not improve predictive performance or if added parameters are not identifiable from the data.

Rubric (1 criterion)

C1. Identifies lack of predictive improvement and/or poor identifiability of added parameters as evidence that extra complexity is unnecessary.

SUP-A11 Supervisor-supplied

Question

Why is a normal distribution used for the initial parasite age distribution?

Valid reference answer

It is a simple way to represent initial synchrony or asynchrony with only two parameters, the mean age and its spread.

Rubric (1 criterion)

C1. Explains that a normal age distribution compactly represents initial synchrony or asynchrony through a mean and spread.

SUP-A12 Supervisor-supplied

Question

How does a lower parasite multiplication factor (PMF) change the predicted parasite trajectory?

Valid reference answer

It reduces the rate of exponential expansion, so parasite densities rise more slowly and peak later or at lower levels.

Rubric (1 criterion)

C1. Explains that a lower PMF reduces cycle-to-cycle expansion and therefore produces slower growth and/or a later or lower peak.

SUP-A13 Supervisor-supplied

Question

A within-host model predicts smooth exponential growth but fails to reproduce the oscillations observed in microscopy parasite count data. What biological process is most likely missing?

Valid reference answer

The model is probably missing explicit parasite age structure and sequestration. Without representing the developmental stages of the parasite and sequestration of mature stages, peripheral parasite densities will not oscillate.

Rubric (2 criteria)

C1. Identifies explicit parasite age structure and sequestration as the missing biological representation.

C2. Explains that stage progression, mature-stage disappearance, and reappearance after rupture generate peripheral oscillations.

SUP-A14 Supervisor-supplied

Question

Why is an age-structured model generally better able to reproduce parasite count fluctuations than a simple exponential growth model?

Valid reference answer

Because an age-structured model tracks parasites through their developmental stages, allowing mature parasites to sequester and young parasites to reappear after schizont rupture. A simple exponential model treats all parasites identically and therefore cannot generate periodic oscillations.

Rubric (2 criteria)

C1. Explains that age structure tracks stage progression, sequestration of mature parasites, and reappearance of young parasites after rupture.

C2. Contrasts this with a simple exponential model in which all parasites behave identically and periodic peripheral oscillations are not generated.

SELF-A01 Self-authored

Question

Can female-gametocyte density estimated from Pfs25 RT-qPCR be interpreted as a direct measure of mosquito infectivity? Give one assay-related and one biological reason for your answer.

Valid reference answer

No. Pfs25 RT-qPCR is an assay-dependent transcript measurement and does not establish that detected gametocytes are viable or fertilisation-competent. Mosquito infection also depends on male-gametocyte availability, sex ratio, nonlinear density effects, and host or mosquito factors.

Rubric (3 criteria)

C1. Rejects interpreting female Pfs25 RT-qPCR density as a direct measure of mosquito infectivity.

C2. Gives an assay-related reason, such as transcript calibration or failure to establish viability or fertilisation competence.

C3. Gives a biological reason, such as dependence on male gametocytes, sex ratio, nonlinear density effects, or host and mosquito factors.

SELF-A02 Self-authored

Question

An assay reports a nondetect below 0.02 parasites per microlitre. Assuming uniform distribution in 5 L of blood, what parasite count corresponds to this limit? If F_Y is the observation-model CDF, what is the likelihood contribution of the nondetect?

Valid reference answer

The limit corresponds to 0.02 x 5 x 10^6 = 10^5 parasites. The observation is left-censored at L, so its likelihood contribution is Pr(Y < L) = F_Y(L), not a point observation of zero parasites.

Rubric (2 criteria)

C1. Correctly converts 0.02 parasites per microlitre in 5 litres of blood to 100,000 parasites.

C2. Treats the nondetect as left-censored and gives the likelihood contribution as F_Y(L), or an equivalent probability below the detection limit, rather than a point observation of zero.

SELF-A03 Self-authored

Question

Construct a minimal hourly age-structured model of the 48-hour asexual blood-stage cycle. Include parasite ageing, survival, schizont rupture, total burden, and an age-dependent mapping to peripheral parasite density.

Valid reference answer

Let P_a(t) be the host-wide number of infected erythrocytes aged a to a+1 hours, a = 0,...,47. Use P_(a+1)(t+1) = s_a P_a(t) for a < 47 and P_0(t+1) = R s_47 P_47(t), where R is effective establishment per rupturing schizont. Total burden is T(t) = sum_a P_a(t). If v_a is the peripheral fraction at age a and V_b is blood volume, expected peripheral density is A(t) = V_b^(-1) sum_a v_a P_a(t).

Rubric (4 criteria)

C1. Defines parasite abundance in discrete hourly age classes across a 48-hour cycle.

C2. Gives ageing transitions with age-specific survival and a rupture transition that returns viable progeny to age zero through an effective multiplication factor.

C3. Defines total burden by summing all age classes.

C4. Maps latent age classes to peripheral density using an age-dependent visibility or sequestration function and blood-volume scaling.

SELF-A04 Self-authored

Question

Extend an hourly age-structured parasite model to include concentration- and age-dependent antimalarial killing. Define drug survival over one hour and show how it changes the ageing and rupture equations.

Valid reference answer

Let kappa_a(C) be an age-specific killing hazard. Drug survival during hour [t,t+1) is D_a(t) = exp{-integral_t^(t+1) kappa_a[C(u)]du}. The treated transitions are P_(a+1)(t+1) = s_a D_a(t)P_a(t) and P_0(t+1) = R s_47 D_47(t)P_47(t). The PK model, exposure compartment, concentration-effect form, stage specificity, and any delayed injury require empirical validation.

Rubric (3 criteria)

C1. Defines an age- and concentration-dependent killing hazard and corresponding one-hour drug-survival probability.

C2. Applies drug survival to both ordinary ageing transitions and the terminal rupture transition.

C3. Recognises that the pharmacokinetic, concentration-effect, stage-specific, or delayed-injury assumptions require empirical support.

3 Synthesis and critique

SUP-S01 Supervisor-supplied

Question

If microscopy shows oscillations but qPCR is smoother, what does that suggest?

Valid reference answer

The oscillations may be partly due to what the measurement captures, but the smoother qPCR profile can also reflect improved sensitivity and reduced sparsity in detection.

Rubric (2 criteria)

C1. Explains that the contrast may arise partly from assay-specific observation processes.

C2. Recognises that smoother qPCR trajectories can reflect greater sensitivity or fewer sparse nondetects.

SUP-S02 Supervisor-supplied

Question

How would you distinguish a true biological plateau from a detection-limited plateau?

Valid reference answer

By using additional data such as qPCR, stage-specific counts, or repeated measures that extend beyond the microscopy detection range.

Rubric (1 criterion)

C1. Proposes additional measurements that can distinguish biology from a detection floor, such as qPCR, stage-specific counts, or repeated observations extending beyond the microscopy range.

SUP-S03 Supervisor-supplied

Question

Why is it risky to assign one specific mechanism to a fitted growth-limiting term?

Valid reference answer

Different processes such as immunity, resource limitation, sexual commitment, and density dependence can produce similar parasite count patterns.

Rubric (1 criterion)

C1. Explains that several mechanisms can produce similar fitted growth limitation, for example immunity, resource limitation, sexual commitment, or density dependence.

SUP-S04 Supervisor-supplied

Question

How would you decide whether to model asexual parasites and gametocytes jointly?

Valid reference answer

Do so when gametocyte data or transmission outcomes are available and the scientific question includes sexual differentiation or infectivity.

Rubric (1 criterion)

C1. Links joint asexual-gametocyte modelling to the availability of gametocyte or transmission data and to a scientific question involving sexual differentiation or infectivity.

SUP-S05 Supervisor-supplied

Question

What additional data would help separate host-mediated killing from parasite-intrinsic regulation?

Valid reference answer

Immune markers, gametocyte counts, sequestration markers, and possibly intervention or perturbation data.

Rubric (1 criterion)

C1. Proposes measurements capable of separating host killing from parasite regulation, such as immune, gametocyte, sequestration, intervention, or perturbation data.

SUP-S06 Supervisor-supplied

Question

How would changing the assumed sequestration age affect fitted trajectories?

Valid reference answer

It would change how long parasites remain visible in peripheral blood, shifting the timing and amplitude of the predicted oscillations.

Rubric (2 criteria)

C1. Explains that changing sequestration age changes how long parasites remain peripherally visible.

C2. Connects this change to the timing and/or amplitude of predicted oscillations.

SUP-S07 Supervisor-supplied

Question

Why might a model that fits the training data well still fail on new trajectories?

Valid reference answer

It may overfit idiosyncratic features of the training set or miss the real biological process that drives new data.

Rubric (1 criterion)

C1. Identifies overfitting to training-specific features and/or omission of the true generalising biological process as causes of poor performance on new trajectories.

SUP-S08 Supervisor-supplied

Question

What is the advantage of keeping the latent biology separate from the observation process?

Valid reference answer

It lets the model distinguish parasite dynamics from measurement artifacts and makes the biological assumptions easier to test.

Rubric (1 criterion)

C1. Explains that separating latent biology from observation distinguishes biological dynamics from measurement effects and makes assumptions independently testable.

SUP-S09 Supervisor-supplied

Question

How would you adapt a model if the goal changed from parasite counts to transmission prediction?

Valid reference answer

Add sexual-stage dynamics and a link from gametocyte density to mosquito infectivity.

Rubric (2 criteria)

C1. Adds sexual-stage or gametocyte dynamics.

C2. Adds a mapping from gametocyte density or state to mosquito infectivity or transmission.

SUP-S10 Supervisor-supplied

Question

Why is a parsimonious model often preferable when the data are sparse?

Valid reference answer

It is easier to identify, less prone to overfitting, and more likely to generalize to new data.

Rubric (1 criterion)

C1. Explains that parsimony improves identifiability and reduces overfitting when observations are sparse, supporting generalisation.

SUP-S11 Supervisor-supplied

Question

Can daily fluctuations in parasite count profiles alone be considered evidence for immune responses?

Valid reference answer

No. Daily fluctuations are primarily explained by parasite age structure and sequestration. Immune responses generally influence the longer-term trajectory (attenuation, plateau, decline) rather than the periodic oscillations themselves.

Rubric (3 criteria)

C1. Rejects daily fluctuations alone as evidence of an immune response.

C2. Attributes short-period oscillations primarily to parasite age structure and sequestration.

C3. Distinguishes these oscillations from longer-term attenuation, plateau, or decline that could reflect immune control.

SUP-S12 Supervisor-supplied

Question

A model accurately reproduces the 48-hour oscillations in parasite density but consistently overestimates parasite counts after day 8. What does this suggest about the model?

Valid reference answer

The model captures parasite age structure and sequestration correctly but lacks a mechanism that limits parasite expansion later in infection, such as immune responses, resource limitation, density-dependent regulation, or sexual commitment.

Rubric (2 criteria)

C1. Infers that the model captures age structure and sequestration because it reproduces the approximately 48-hour oscillations.

C2. Infers that it lacks a later growth-limiting mechanism, such as immunity, resource limitation, density dependence, or sexual commitment.

SELF-S01 Self-authored

Question

Extend an age-structured asexual model with sexual commitment, delayed immature-gametocyte development, and circulating mature female and male gametocytes. State whether commitment follows a same-cycle or next-cycle convention and conserve parasite flow.

Valid reference answer

Under a next-cycle convention, if Q(t) is viable rupture progeny, send B(t) = c(t)Q(t) into a delayed immature-gametocyte chain and (1-c(t))Q(t) into the next asexual generation. A transit chain with rate k = m/T_g gives mean maturation time T_g, defensibly about 8-12 days. Split mature outflow into female and male compartments with their own loss rates. Under a same-cycle convention, divert a pre-rupture asexual flow instead; it must not include the rupture multiplier. Mature gametocytes do not replicate in human blood.

Rubric (4 criteria)

C1. States a same-cycle or next-cycle commitment convention and diverts committed parasites at the corresponding point without double counting.

C2. Conserves progeny flow between asexual renewal and the committed sexual pathway.

C3. Represents delayed immature-gametocyte development, for example with a transit chain having an approximately 8-12 day mean delay.

C4. Splits mature outflow into female and male compartments with appropriate loss and no replication in human blood.

SELF-S02 Self-authored

Question

How should a within-host model link latent asexual, female-gametocyte, and male-gametocyte states to microscopy, pan-Plasmodium DNA qPCR, and sex-enriched RT-qPCR observations?

Valid reference answer

Use a separate observation function and error model for each assay. Microscopy should depend on the peripheral, age-weighted asexual state and sampled blood volume. Pan-Plasmodium qPCR needs target-specific scaling and may include DNA from multiple detectable stages. Female- and male-enriched RT-qPCR require separate calibration and may include cross-expression. A nondetect below L contributes Pr(Y < L), and molecular positivity alone does not establish viability or infectivity.

Rubric (4 criteria)

C1. Uses assay-specific observation functions and error models rather than one shared mapping.

C2. Maps microscopy to the peripheral age-weighted asexual state and an appropriate density scale.

C3. Allows pan-Plasmodium DNA qPCR and sex-enriched RT-qPCR to have target-specific calibration and stage or cross-expression contributions.

C4. Treats nondetects as censored observations and does not equate molecular positivity with viability or infectivity.

SELF-S03 Self-authored

Question

A treatment model contains unknown initial biomass, parasite-age distribution, multiplication, sequestration, stage-specific killing, drug exposure, and immune clearance. Only peripheral parasite density is measured every six hours for 48 hours. Which effects are confounded, and what additional measurements would help separate them?

Valid reference answer

Initial biomass is confounded with the circulating fraction determined by age and sequestration. Stage-specific killing is confounded with unmeasured exposure and the stages present at treatment. Sequestration can mimic clearance, while multiplication, drug killing, and immune clearance contribute only a net observed rate. Stage-resolved counts, measured pharmacokinetics, an independent biomass marker such as PfHRP2, longer untreated observations, and immune measurements would reduce these ambiguities. Exact structural identifiability cannot be claimed without the complete equations.

Rubric (5 criteria)

C1. Identifies confounding between initial biomass, parasite age distribution, and the fraction visible because of sequestration.

C2. Identifies confounding among stage-specific killing, unmeasured drug exposure, and parasite stages present at treatment.

C3. Explains that sequestration can resemble clearance and that multiplication, drug killing, and immune clearance may be observed only through a net rate.

C4. Proposes informative additional data such as stage-resolved counts, pharmacokinetics, an independent biomass marker, longer untreated follow-up, or immune measurements.

C5. Avoids claiming exact structural identifiability without complete model equations.

SELF-S04 Self-authored

Question

Why is parasite-clearance half-life usually more informative about reduced artemisinin susceptibility than time to microscopy negativity, and why is it not a pure parasite-intrinsic killing rate?

Valid reference answer

Time to negativity depends on starting density, the microscopy detection limit, and sampling frequency. During log-linear decline P(t) = P0 exp(-kt), the half-life ln(2)/k is independent of P0 and is therefore more comparable. It still measures disappearance from peripheral blood, not parasite death alone, and is affected by parasite age, sequestration, schizogony, drug exposure, splenic clearance, immunity, and measurement choices.

Rubric (4 criteria)

C1. Explains that time to microscopy negativity depends on starting density, detection limit, and/or sampling frequency.

C2. Explains why a log-linear clearance half-life is more comparable and gives or correctly interprets ln(2)/k.

C3. States that peripheral clearance half-life is not a pure parasite-intrinsic killing rate.

C4. Identifies relevant influences such as parasite age, sequestration, schizogony, exposure, splenic or immune clearance, or measurement choices.

4 Computational modelling

SELF-C01 Self-authored

Question

At integer hour t, P_a(t) is the number of infected erythrocytes in parasite-age bin a, for a = 0,...,47. Initially P_0 = 120, P_12 = 80, P_25 = 50, P_47 = 30, and all other bins are zero. Apply the simultaneous update P_(a+1)(t+1) = 0.98 P_a(t) for a = 0,...,46 and P_0(t+1) = 8 x 0.98 P_47(t). Define peripheral count A(t) = sum_(a=0)^25 P_a(t) and total burden T(t) = sum_(a=0)^47 P_a(t). Without intermediate rounding, calculate A(24), A(48), T(48), A(96), and T(96). Report two decimal places.

Valid reference answer

A(24) = 467.99; A(48) = 758.37; T(48) = 849.38; A(96) = 2300.51; T(96) = 2576.57.

Rubric (5 criteria)

C1. Reports A(24) = 467.99.

C2. Reports A(48) = 758.37.

C3. Reports T(48) = 849.38.

C4. Reports A(96) = 2300.51.

C5. Reports T(96) = 2576.57.

SELF-C02 Self-authored

Question

Discretise parasite age as a = 0,...,47 hours and time as t_n = 0.25n hours. Set w_a = exp[-(a-8)^2/(2 x 4^2)] and p_a^0 = 10^6 w_a/sum_(b=0)^47 w_b. At step n set C_n = 200 x 2^(-t_n/6) and kappa_(a,n) = h_a C_n/(50+C_n), where h_a = 0.15 h^-1 for 18 <= a <= 40 and 0.02 h^-1 otherwise. Define q_0^n = 8p_47^n and q_a^n = p_(a-1)^n for a = 1,...,47, then update all bins simultaneously by p_a^(n+1) = p_a^n + 0.25[q_a^n - p_a^n - kappa_(a,n)p_a^n]. With A_n = sum_(a=0)^25 p_a^n and T_n = sum_(a=0)^47 p_a^n, calculate A at 24, 48, and 72 hours and T at 72 hours. Iterate exactly as specified without intermediate rounding and report one decimal place.

Valid reference answer

A(24) = 80992.6; A(48) = 2272713.5; A(72) = 1245629.8; T(72) = 3159203.8.

Rubric (4 criteria)

C1. Reports A(24) = 80992.6.

C2. Reports A(48) = 2272713.5.

C3. Reports A(72) = 1245629.8.

C4. Reports T(72) = 3159203.8.

SELF-C03 Self-authored

Question

Model log10 parasite density at day t_i as Y_i ~ Normal(mu_i, 0.15^2), where mu_i = alpha + r t_i. The observations (t_i, Y_i) are (0, ND), (1, ND), (2, -0.72), (3, -0.38), (4, 0.05), (5, 0.29), and (6, 0.67); ND means Y_i < -1. Let phi and Phi be the standard-normal density and CDF. Maximise ell(alpha,r) = sum_detected log{phi[(Y_i-mu_i)/0.15]/0.15} + sum_ND log Phi[(-1-mu_i)/0.15], subject to -3 <= alpha <= 1 and 0 <= r <= 1. Report alpha, r, the maximised log-likelihood, and the time (-1-alpha)/r at which the latent mean crosses the detection limit, each to three decimal places.

Valid reference answer

alpha = -1.478; r = 0.362 per day; maximised log-likelihood = 4.419; detection-limit crossing time = 1.320 days.

Rubric (4 criteria)

C1. Reports alpha = -1.478.

C2. Reports r = 0.362 per day.

C3. Reports the maximised log-likelihood as 4.419.

C4. Reports the detection-limit crossing time as 1.320 days.

SELF-C04 Self-authored

Question

Consider the Gaussian hierarchical model y_ij ~ Normal(theta_i, 0.2^2), theta_i ~ Normal(mu, 0.3^2), and mu ~ Normal(0, 1^2), where the values after each comma are standard deviations. The three groups are y_1 = (-0.1, 0.0, 0.1), y_2 = (0.8, 0.9, 1.0), and y_3 = (1.8, 2.0, 2.2). Compute the exact joint Gaussian posterior. For mu, theta_1, theta_2, and theta_3, report the posterior mean and the central interval defined as mean +/- 1.96 posterior standard deviations, each to three decimal places.

Valid reference answer

mu: 0.934 [0.577, 1.292]; theta_1: 0.121 [-0.096, 0.337]; theta_2: 0.904 [0.688, 1.121]; theta_3: 1.863 [1.646, 2.079].

Rubric (4 criteria)

C1. Reports the posterior for mu as mean 0.934 with interval [0.577, 1.292].

C2. Reports the posterior for theta_1 as mean 0.121 with interval [-0.096, 0.337].

C3. Reports the posterior for theta_2 as mean 0.904 with interval [0.688, 1.121].

C4. Reports the posterior for theta_3 as mean 1.863 with interval [1.646, 2.079].

SELF-C05 Self-authored

Question

For 12 observations, the pointwise PSIS-LOO expected log predictive densities are A = (-1.2, -0.8, -1.5, -2.0, -0.9, -1.1, -3.2, -0.7, -1.8, -1.0, -2.4, -1.3) and B = (-1.5, -1.0, -1.4, -2.5, -1.0, -1.3, -2.8, -0.9, -2.1, -1.2, -2.0, -1.6). Let d_i = A_i-B_i, Delta = sum_i d_i, and SE(Delta) = sqrt(12 s_d^2), where s_d^2 = sum_i(d_i-d_bar)^2/11. The Pareto-k values are k_A = (0.20, 0.30, 0.10, 0.40, 0.50, 0.20, 0.82, 0.30, 0.60, 0.20, 0.75, 0.40) and k_B = (0.25, 0.35, 0.20, 0.45, 0.40, 0.25, 0.65, 0.30, 0.55, 0.20, 0.80, 0.45). Calculate Delta and its SE to three decimal places, count observations for which either model has k > 0.7, and state the justified model-comparison conclusion.

Valid reference answer

Delta = 1.400 and SE(Delta) = 0.967, favouring model A by about 1.45 standard errors. Two observations have k > 0.7 in at least one model. The difference is not strong relative to its uncertainty, and the problematic Pareto-k values make the PSIS-LOO comparison unreliable without exact leave-one-out refits or further investigation.

Rubric (4 criteria)

C1. Reports Delta = 1.400 in favour of model A.

C2. Reports SE(Delta) = 0.967.

C3. Identifies two observations for which at least one model has Pareto-k greater than 0.7.

C4. Concludes that the ELPD difference is weak relative to its uncertainty and that the Pareto-k diagnostics warrant exact leave-one-out refits or further investigation.